

ORIGINAL ARTICLE

Strategic value of minimally invasive LVAD implantation in an emerging national program: A comparative cohort study

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ABSTRACT

Background and Aim: Minimally invasive surgical (MIS) approaches for left ventricular assist device (LVAD) implantation have demonstrated favorable outcomes in high-volume centers. However, evidence supporting their effectiveness in emerging mechanical circulatory support (MCS) programs remains limited. This study aimed to evaluate whether minimally invasive LVAD implantation is associated with reduced postoperative morbidity compared with conventional median sternotomy within an emerging national LVAD program.

Research design and Methods: We conducted a retrospective observational cohort study including 60 consecutive adult patients who underwent durable continuous-flow LVAD implantation between 2020 and 2025 at a tertiary referral center. Patients were treated either using a minimally invasive thoracotomy-based approach (MIS, n = 30) or conventional median sternotomy (n = 30). The primary outcome was a composite postoperative morbidity endpoint, including reoperation for bleeding, right ventricular dysfunction, neurological events, acute kidney injury requiring renal replacement therapy, and clinically significant infection. Secondary outcomes included perioperative parameters, ventricular remodeling, functional recovery, and survival.

Results: Baseline demographic and clinical characteristics were comparable between groups. The MIS approach was associated with a significantly shorter cardiopulmonary bypass time (median 120 vs. 181 minutes; p = 0.028), although intraoperative blood loss was higher. The composite postoperative morbidity rate was significantly lower in the MIS group compared with sternotomy (30% vs. 50%; p = 0.04). Both surgical approaches



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resulted in substantial functional improvement and reverse left ventricular remodeling. One-year survival did not differ significantly between groups (log-rank $p = 0.67$). On multivariable analysis, MIS implantation was independently associated with a lower risk of postoperative complications (OR 0.42, 95% CI 0.18-0.96).

Conclusions: In an emerging LVAD program, minimally invasive implantation was associated with a significantly lower postoperative morbidity burden and comparable survival. These findings support the strategic adoption of minimally invasive LVAD implantation in appropriately selected patients within developing MCS programs. (www.actabiomedica.it)

Key words: left ventricular assist device, minimally invasive surgery, mechanical circulatory support, thoracotomy, postoperative morbidity, emerging cardiac surgery programs, resource-limited settings

Introduction

Advanced heart failure represents a growing global health burden, with a disproportionate impact on middle-income countries where access to heart transplantation is limited and donor availability remains low. In these settings, durable left ventricular assist devices (LVADs) have evolved from a bridging therapy to a cornerstone treatment for end-stage heart failure. As national LVAD programs expand, clinical priorities increasingly extend beyond survival alone toward reducing postoperative morbidity, optimizing resource utilization, and ensuring long-term program sustainability.

Heart failure (HF) is a complex clinical syndrome caused by structural and/or functional cardiac abnormalities and remains a leading cause of morbidity, mortality, and healthcare utilization worldwide (1). According to the Global Burden of Disease (GBD) 2019 study, the global prevalence of HF exceeded 64 million individuals, with 31.8 million cases reported in Asia and Oceania (95% uncertainty interval [UI]: 25.94-39.25 million), corresponding to an age-standardized prevalence rate of 722.45 per 100,000 population (95% UI: 591.97-891.64) (2,3). Marked regional heterogeneity exists within Asia, with age-standardized prevalence ranging from 389.97 per 100,000 population in South Asia to 1,014 per 100,000 population in East Asia; Central Asia demonstrates an intermediate burden of 544.18 per 100,000 population, indicating a substantial population at risk of progression to advanced HF (4).

In Kazakhstan, national registry data indicate a substantial HF burden, with more than 526,000 adults diagnosed between 2014 and 2019 (5). Despite a decline in HF incidence, all-cause mortality among patients with HF increased nearly threefold during this period, from 356 to 975 deaths per million population, accounting for 2.36 million disability-adjusted life years (DALYs) in 2019, predominantly due to premature mortality (5,6). This epidemiological pattern is consistent with that observed in other countries with emerging advanced HF programs, underscoring the growing number of patients progressing to stages where guideline-directed medical therapy is no longer sufficient (7).

For patients with advanced HF refractory to medical therapy, durable LVADs represent a key life-prolonging treatment option, either as destination therapy or as a bridge to further treatment (8,9). The national mechanical circulatory support program in Kazakhstan has been developing since 2012, with 493 LVAD implantations performed nationwide between 2012 and 2023, reflecting increasing regional experience with this technology (10). Recent national evidence further supports the effectiveness of LVAD therapy, with a retrospective cohort study of 100 recipients demonstrating significant and sustained improvements in health-related quality of life at 1, 6, and 12 months after implantation, as assessed by the EQ-5D-5L questionnaire (11).

As LVAD programs expand, attention has increasingly focused on optimizing surgical implantation

strategies (12). Conventional LVAD implantation via full median sternotomy provides reliable exposure but is associated with substantial surgical trauma, perioperative bleeding, and prolonged recovery (13,14). Advances in pump miniaturization and surgical techniques have enabled minimally invasive, predominantly thoracotomy-based approaches aimed at reducing perioperative morbidity while maintaining effective ventricular unloading (15,16). A systematic review and meta-analysis of 25 observational studies including 3,072 patients demonstrated that thoracotomy-based LVAD implantation was associated with significantly lower perioperative morbidity than median sternotomy, including reduced blood transfusion requirements, fewer reoperations for bleeding (odds ratio 0.34, 95% CI: 0.22-0.54), lower rates of postoperative right ventricular assist device implantation (odds ratio 0.53, 95% CI: 0.36-0.77), and shorter intensive care unit and hospital length of stay, without significant differences in adjusted perioperative or overall mortality (17).

Despite these favorable findings, most available data originate from high-volume centers with mature LVAD programs, and comparative evidence from emerging mechanical circulatory support programs remains limited (18). In resource-constrained healthcare systems such as Kazakhstan, reducing perioperative complications and accelerating postoperative recovery may have a particularly important impact on clinical outcomes, resource utilization, and program sustainability (19).

Therefore, this study aimed to evaluate whether minimally invasive LVAD implantation is associated with reduced postoperative morbidity compared with conventional median sternotomy within an emerging national LVAD program, and to explore its implications for surgical strategy and program development.

Patients and methods

Study design, setting, and ethical considerations

This study was designed as a retrospective observational cohort study. The analysis was based on routinely collected clinical data from patients who underwent

durable left ventricular assist device implantation at the Research and Clinical Center for Cardiac Surgery and Transplantology (RCCCST), Taraz, Kazakhstan, a tertiary referral center for advanced heart failure and mechanical circulatory support.

All LVAD implantations were performed between 1 January 2020 and 1 September 2025. The study was conducted in accordance with the principles of the Declaration of Helsinki and was reviewed and approved by the Local Ethics Committee of Kazakhstan's Medical University "KSPH" (protocol #4 from 21.05.2025). Given the retrospective nature of the study and the use of anonymized data obtained from existing medical records, written informed consent for study participation was waived by the Ethics Committee. Nevertheless, all patients had provided standard written informed consent for surgical intervention and perioperative data use at the time of hospitalization, in accordance with national regulations.

This study represents a real-world consecutive cohort reflecting routine clinical practice in an emerging mechanical circulatory support program. All eligible patients undergoing durable LVAD implantation during the study period were included without exclusion based on surgical approach feasibility. This design was chosen to minimize selection bias related to case inclusion and to capture the full institutional experience.

Study population and sampling strategy

A consecutive sampling strategy was applied. All adult patients who underwent durable LVAD implantation at RCCCST during the study period were screened for eligibility. No a priori sample size calculation was performed, as the study aimed to include the entire national single-center experience during the specified timeframe.

A total of 60 consecutive patients met the eligibility criteria and had complete perioperative and follow-up data available; all were included in the final analysis. Patients were stratified into two groups according to the surgical implantation strategy: 1) Minimally invasive surgical approach (MICS); 2) Conventional median sternotomy. This approach minimized selection bias related to case inclusion and ensured comprehensive representation of institutional practice.

All procedures were performed by a dedicated multidisciplinary mechanical circulatory support team, consisting of experienced cardiac surgeons, cardiac anesthesiologists, perfusionists, intensivists, and specialized nursing staff. The core surgical team remained consistent throughout the study period, ensuring procedural standardization and reducing operator-dependent variability. Perioperative management protocols, including anesthesia, cardiopulmonary bypass management, anticoagulation, and postoperative intensive care, were standardized and applied uniformly to both study groups.

Eligibility criteria

INCLUSION CRITERIA

Patients were eligible for inclusion if they met all of the following criteria: age 18 to 75 years at the time of surgery; diagnosis of advanced heart failure refractory to guideline-directed medical therapy; INTERMACS functional profile III or IV at the time of LVAD implantation; implantation of a durable continuous-flow LVAD; availability of complete perioperative data and follow-up information.

Patients were included regardless of sex, ethnicity, or heart failure etiology.

EXCLUSION CRITERIA

Patients were excluded if any of the following conditions were present: contraindications to durable mechanical circulatory support as defined by national and international guidelines; irreversible end-organ dysfunction (advanced hepatic, renal, or neurological failure) not expected to improve with LVAD support; active systemic infection or sepsis at the time of implantation; uncontrolled coagulopathy or active bleeding; incomplete clinical records or insufficient follow-up data precluding reliable outcome assessment.

The choice of surgical approach (minimally invasive versus median sternotomy) was not randomized and was determined on a case-by-case basis. Decision-making was guided by a combination of: patient-specific anatomical characteristics (chest anatomy, left ventricular apex accessibility, ascending

aorta position); results of preoperative imaging, including computed tomography; presence of prior cardiac surgery or mediastinal adhesions; severity of right ventricular dysfunction and overall clinical status; institutional experience and intraoperative feasibility. Importantly, all patients were considered eligible for either approach, and no formal exclusion from a specific surgical strategy was applied. This reflects real-world clinical decision-making in an emerging LVAD program and allows comparison of outcomes between approaches under routine practice conditions.

Surgical technique

LVAD IMPLANTATION VIA MEDIAN STERNOTOMY

In the conventional surgical group, LVAD implantation was performed via full median sternotomy. After standard preparation, the sternum was divided, and the pericardium was opened longitudinally (approximately 10-12 cm) to expose the heart and great vessels (Figure 1).

All procedures were conducted under cardiopulmonary bypass (CPB). Arterial cannulation was established in the ascending aorta using a 20-22 Fr cannula, with venous drainage via the right atrium or bicaval cannulation (28-32 Fr). Normothermic perfusion (36-37 °C) was maintained with a target flow of 2.2-2.5 L/min/m². The left ventricular apex was identified using anatomical landmarks and transesophageal echocardiography. An apical sewing ring was secured using 8-12 pledgeted polypropylene sutures, followed by a ventriculotomy measuring approximately 1.5-2.0 cm. The inflow cannula was inserted and oriented toward the mitral valve, parallel to the interventricular septum under echocardiographic guidance. The outflow graft was anastomosed end-to-side to the ascending aorta at an angle of approximately 40-45° using a continuous 5/0 polypropylene suture, with careful deairing.

After LVAD activation, gradual ventricular unloading was achieved while weaning from CPB under hemodynamic and echocardiographic monitoring. Following confirmation of stable device function, CPB was discontinued, hemostasis achieved, mediastinal drains placed, and the sternum was closed using standard wire fixation.

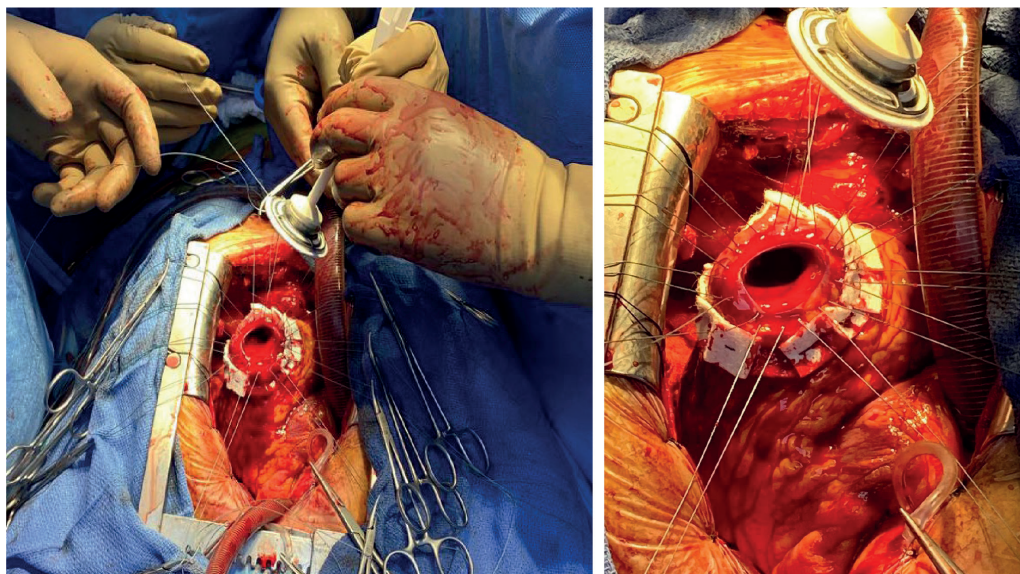


Figure 1. Traditional median sternotomy for LVAD implantation.

Minimally invasive LVAD implantation

In the minimally invasive surgical group, LVAD implantation was performed using a thoracotomy-based approach. Patients were positioned supine with a 20–30° rightward torso rotation and elevation of the left hemithorax to optimize exposure of the left ventricular apex.

CPB was established predominantly via peripheral cannulation, with venous drainage through the femoral vein (21–25 Fr) and arterial inflow via the femoral artery (15–19 Fr) or, when feasible, via limited ascending aortic access. Normothermic perfusion (36–37 °C) was maintained with a target flow of 2.2–2.4 L/min/m², and systemic anticoagulation was achieved to maintain an activated clotting time >480 s. Primary access was achieved through a left anterolateral mini-thoracotomy measuring approximately 6–8 cm in the fifth or sixth intercostal space. The pericardium was opened longitudinally over 5–6 cm parallel to the phrenic nerve. Access for the outflow graft anastomosis was obtained either via an upper J-shaped ministernotomy (5–6 cm) or a right parasternal mini-thoracotomy, depending on preoperative computed tomography findings and ascending aortic anatomy (Figure 2).

The inflow cannula implantation site was selected on the anterolateral apical surface under echocardiographic guidance. The sewing ring was secured using 10–12 pledgeted 2/0 polypropylene sutures, followed by a ventriculotomy measuring 1.5–2.0 cm. The inflow cannula was oriented toward the mitral valve, parallel to the interventricular septum, and the outflow graft was anastomosed end-to-side to the ascending aorta at an angle of approximately 45° using a continuous 5/0 polypropylene suture.

LVAD support was initiated at low rotational speed with gradual reduction of CPB. Separation from CPB was performed stepwise under continuous hemodynamic and echocardiographic monitoring. After confirmation of stable device function, hemostasis was achieved, pleural and pericardial drains were placed, incisions were closed in layers, and patients were transferred to the intensive care unit (Figure 3).

Details of perioperative management are provided in Supplementary Table S1.

Data collection and definitions

Data were obtained from electronic medical records, operative reports, intensive care unit charts, and standardized echocardiography protocols. Baseline



Figure 2. The minimally invasive approach for LVAD implantation.



Figure 3. Completion of LVAD device placement surgery using minimally invasive access.

demographic and clinical data, including age, sex, body mass index, heart failure etiology, INTERMACS profile, and history of prior cardiac surgery, were extracted. Preoperative laboratory parameters (serum creatinine, total bilirubin, and NT-proBNP) were

obtained within 24-48 hours prior to LVAD implantation and analyzed as baseline values. Baseline functional capacity was assessed preoperatively using the six-minute walk test (6MWT). Long-term functional status was assessed at 12 months postoperatively using

the New York Heart Association (NYHA) functional classification. Transthoracic echocardiography was performed: preoperatively (baseline); at 1, 3, and 6 months after LVAD implantation. Echocardiographic parameters included left ventricular end-diastolic diameter (LVEDD), left ventricular end-systolic diameter (LVESD), right ventricular fractional area change (RV FAC, %), and tricuspid annular plane systolic excursion (TAPSE, cm). Serial changes were analyzed to assess reverse ventricular remodeling.

The primary outcome was a composite postoperative morbidity endpoint, defined as the occurrence of any major postoperative complication, including reoperation for bleeding, right ventricular dysfunction requiring inotropic or mechanical support, neurological events, acute kidney injury requiring renal replacement therapy, or clinically significant infection.

Secondary outcomes included intraoperative blood loss, cardiopulmonary bypass duration, duration of mechanical ventilation, length of intensive care unit and hospital stay, echocardiographic measures of ventricular remodeling, functional status at 12 months, and overall survival. The composite postoperative morbidity endpoint is defined in Supplementary Table S2.

Statistical analysis

Continuous variables were assessed for normality and are presented as mean $\pm$ standard deviation or median with interquartile range [IQR], as appropriate. Categorical variables are presented as counts and percentages. Comparisons between groups were performed using: student's t-test or Mann-Whitney U test for continuous variables; chi-square or Fisher's exact test for categorical variables; survival was analyzed using the Kaplan-Meier method and compared with the log-rank test.

Multivariable logistic regression analysis was conducted to identify independent predictors of postoperative complications. Given the significant baseline difference in right ventricular systolic function between the study groups, a sensitivity analysis stratified by baseline RV function was performed to assess the robustness of the association between surgical approach and postoperative complications. A two-sided p-value < 0.05 was considered statistically significant.

Results

Baseline characteristics of study population

A total of 60 patients undergoing durable LVAD implantation were included in the analysis. Of these, 30 patients underwent implantation via a minimally invasive surgical approach, and 30 patients underwent implantation via conventional median sternotomy. All patients had complete perioperative and follow-up data available for analysis. No patients were excluded due to missing key outcome variables. Baseline demographic and clinical characteristics are presented in Table 1.

There were no statistically significant differences with respect to age, sex distribution, body mass index, heart failure etiology, INTERMACS functional profile, or history of prior cardiac surgery.

The vast majority of patients in both cohorts were classified as INTERMACS III-IV, reflecting advanced but hemodynamically stable heart failure at the time of LVAD implantation, with only isolated INTERMACS II cases observed. The distribution of LVAD implantation indications (bridge to transplantation, bridge to candidacy, and destination therapy) was comparable between groups, indicating no relevant baseline imbalance in treatment strategy.

Preoperative risk profile and functional status

All patients in both groups were classified as high surgical risk according to the EuroSCORE system, reflecting advanced disease severity at the time of LVAD implantation. Preoperative laboratory parameters and functional status were assessed within the standard preoperative evaluation window and were considered representative of baseline clinical condition.

Baseline renal function was comparable between groups. Median serum creatinine levels did not differ significantly between patients undergoing minimally invasive LVAD implantation and those treated via conventional sternotomy, indicating the absence of clinically relevant differences in preoperative renal reserve.

Markers of hepatic congestion showed greater variability. Total bilirubin levels tended to be higher

Table 1. Baseline characteristics of patients undergoing LVAD implantation, (median [IQR]).

Characteristic	MICS (n = 30)	Sternotomy (n = 30)	p-value
Age, years	61.0 [52.0-64.0]	57.5 [49.0-61.0]	0.31
Male sex, n (%)	25 (83.3%)	27 (90.0%)	0.46
BMI, kg/m ²	27.1 [24.8-30.2]	28.0 [25.1-31.0]	0.48
Heart failure etiology, n (%)			0.60
Dilated cardiomyopathy	16 (53.3%)	14 (46.7%)	
Ischemic cardiomyopathy	14 (46.7%)	16 (53.3%)	
INTERMACS profile, n (%)			0.95
INTERMACS II	1 (3.3%)	1 (3.3%)	
INTERMACS III	18 (60.0%)	17 (56.7%)	
INTERMACS IV	11 (36.7%)	12 (40.0%)	
Prior cardiac surgery, n (%)	6 (20.0%)	7 (23.3%)	0.75
Indication for LVAD, n (%)			0.63
Bridge to transplantation (BTT)	11 (36.7%)	9 (30.0%)	
Bridge to candidacy (BTC)	7 (23.3%)	6 (20.0%)	
Destination therapy (DT)	12 (40.0%)	15 (50.0%)	

in the sternotomy group; however, the difference did not reach statistical significance, with substantial overlap between interquartile ranges, suggesting a similar overall burden of congestive hepatopathy in both cohorts.

Circulating NT-proBNP concentrations were markedly elevated in all patients, consistent with end-stage heart failure. Median NT-proBNP levels were higher in the MICS group, whereas patients in the sternotomy group demonstrated lower median values with wide interindividual variability, reflecting heterogeneity in neurohormonal activation. The between-group difference approached but did not reach statistical significance.

Baseline functional capacity was severely impaired in both groups. Distances achieved during the six-minute walk test were low overall, reflecting advanced functional limitation prior to LVAD implantation. No statistically significant difference in baseline 6MWT performance was observed between surgical approaches.

A detailed summary of preoperative risk profile, laboratory parameters, and functional status is presented in Table 2.

Intraoperative outcomes

Intraoperative outcomes are summarized in Table 3.

Compared with conventional median sternotomy, the minimally invasive surgical approach was associated with higher intraoperative blood loss. In contrast, cardiopulmonary bypass time was significantly shorter in the MICS group.

The requirement for perioperative blood transfusion occurred less frequently in patients undergoing minimally invasive implantation; however, this difference did not reach statistical significance. The rate of concomitant surgical procedures was comparable between groups. Importantly, no intraoperative conversions from MICS to median sternotomy were observed.

Early postoperative outcomes

Early postoperative outcomes are summarized in Table 4.

Based on verified source data, the duration of mechanical ventilation was longer in the MICS group compared with conventional sternotomy. Length of intensive care unit (ICU) stay and total hospital length of stay did not differ significantly between groups.

Table 2. Preoperative risk profile and functional status, (median [IQR]).

Variable	MICS (n = 30)	Sternotomy (n = 30)	p-value
EuroSCORE category	high risk (100%)	high risk (100%)	—
Creatinine, µmol/L	96.8 [78.0-111.0]	92.1 [74.2-105.0]	0.64
Total bilirubin, µmol/L	14.9 [10.8-22.1]	21.0 [12.7-37.0]	0.10
NT-proBNP, pg/mL	10 010 [5 583-16 307]	4 024 [3 123-7 843]	0.06
6-min walk test, m (baseline)	130 [50-250]	90 [50-150]	0.28

Table 3. Intraoperative outcomes, (median [IQR]).

Outcome	MICS (n = 30)	Sternotomy (n = 30)	p-value
Intraoperative blood loss, mL	400 [363-500]	230 [200-315]	<0.001
Cardiopulmonary bypass time, min	120 [105-139]	181 [119-191]	0.028
Blood transfusion, n (%)	11 (36.7%)	17 (56.7%)	0.12
Concomitant procedures, n (%)	4 (13.3%)	5 (16.7%)	0.72

Table 4. Early postoperative outcomes, (median [IQR]).

Outcome	MICS (n = 30)	Sternotomy (n = 30)*	p-value
Mechanical ventilation, hours	15.0 [10.0-21.0]	10.0 [9.0-13.0]	0.034
ICU stay, days	3.44 [2.69-4.00]	2.96 [2.00-3.96]	0.177
Hospital stay, days	22.0 [17.5-26.75]	21.5 [14.75-31.25]	0.975
30-day mortality, n (%)	1 (3.3%)	2 (6.7%)	0.49

Early (30-day) mortality was low and comparable between surgical approaches. Postoperative complications occurred more frequently in the sternotomy group; however, differences for individual complications did not reach statistical significance.

Postoperative complications occurred significantly more frequently in patients undergoing conventional median sternotomy compared with those treated using a minimally invasive approach. The distribution of complications by surgical technique is presented in Table 5.

Overall, any postoperative complication was observed in half of patients in the sternotomy group, whereas complications occurred in less than one third of patients in the MICS group. Reoperation for bleeding was numerically more frequent after sternotomy,

although this difference did not reach statistical significance.

Other complications, including infectious events, neurological complications, right ventricular dysfunction, and acute kidney injury requiring renal replacement therapy, occurred at low absolute rates. Due to the limited number of events, these complications were analyzed descriptively.

Device-related adverse events are summarized in Supplementary Table S4.

Echocardiographic outcomes

Baseline echocardiographic characteristics are summarized in Table 6. Left ventricular dimensions at baseline were comparable between groups, indicating

Table 5. Postoperative complications according to surgical approach, (median [IQR]).

Complication	MICS (n = 30)	Sternotomy (n = 30)	p-value
Any postoperative complication	9 (30.0%)	15 (50.0%)	0.04
Reoperation for bleeding	2 (6.7%)	6 (20.0%)	0.12
Infectious complications*	3 (10.0%)	5 (16.7%)	0.45
Neurological events (stroke/TIA)	1 (3.3%)	2 (6.7%)	0.55
Right ventricular dysfunction	3 (10.0%)	6 (20.0%)	0.27
Acute kidney injury requiring RRT	0 (0%)	2 (6.7%)	0.15

Table 6. Baseline echocardiographic characteristics, (median [IQR]).

Parameter	MICS (n = 30)	Sternotomy (n = 30)	p-value
RV FAC (%)	39 [35-43]	32 [28-36]	<0.001
LVEDD (cm)	7.2 [6.9-7.5]	7.3 [7.0-7.6]	0.42
LVESD (cm)	6.4 [6.0-6.8]	6.5 [6.1-6.9]	0.51

a similar degree of preoperative ventricular dilatation. In contrast, baseline right ventricular fractional area change (RV FAC) was significantly higher in patients selected for minimally invasive LVAD implantation.

Serial transthoracic echocardiography performed at 1, 3, and 6 months after LVAD implantation demonstrated progressive reverse remodeling of the left ventricle in both cohorts (Table 7). Reductions in LV end-diastolic and end-systolic dimensions were observed at all follow-up time points. Right ventricular fractional area change remained relatively stable during follow-up and was consistently higher in the MICS group.

Changes from Baseline to 6 Months

To quantify the magnitude of ventricular remodeling, within-patient changes from baseline to 6 months were calculated and compared between groups (Table 8). Both surgical approaches resulted in effective left ventricular unloading and reverse remodeling (Figures 4,5).

Minimally invasive implantation was associated with numerically greater reductions in left ventricular dimensions and better preservation of right ventricular systolic function assessed by RV FAC. The reduction in LV end-systolic dimension was significantly greater in the MICS group, whereas differences in LV end-diastolic dimension and RV systolic function did not reach statistical significance.

Follow-Up and survival, multivariable analysis

During the available follow-up period, overall survival was comparable between the two groups. Kaplan-Meier survival analysis demonstrated no statistically significant difference in survival between patients undergoing minimally invasive LVAD implantation and those treated via conventional median sternotomy (log-rank $p = 0.67$). Survival was analyzed using time-to-event methods with censoring at the last available follow-up.

At 12 months after LVAD implantation, the majority of surviving patients in both groups achieved NYHA functional class II, indicating substantial functional improvement following mechanical circulatory support.

A multivariable logistic regression model was constructed to identify independent predictors of postoperative complications (Table 9). After adjustment for age, body mass index, and history of prior cardiac surgery, the minimally invasive surgical approach remained independently associated with a lower risk of postoperative complications. In a sensitivity analysis stratified by baseline right ventricular systolic function, the direction of the association between minimally invasive LVAD implantation and lower postoperative complication rates remained consistent across strata. This suggests that the observed reduction in composite

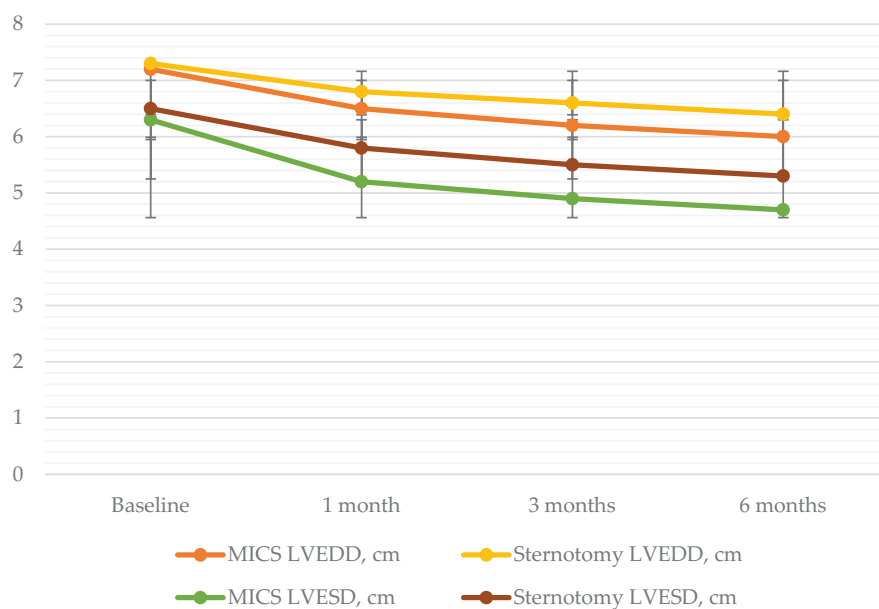


Figure 4. Left ventricular reverse remodeling.

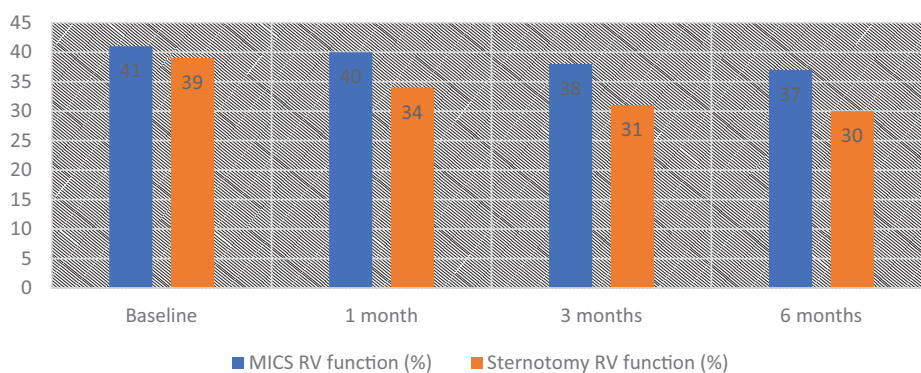


Figure 5. Right ventricular function over time.

Table 7. Echocardiographic follow-up after LVAD implantation, (median [IQR]).

Parameter	Time	MICS	Sternotomy
RV FAC (%)	1 month	40 [36-43]	32 [28-36]
	3 months	37 [34-41]	31 [27-35]
	6 months	36 [33-40]	31 [27-34]
LVEDD (cm)	1 month	6.6 [6.2-7.0]	6.9 [6.5-7.2]
	3 months	6.4 [6.0-6.8]	6.7 [6.4-7.0]
	6 months	6.3 [5.9-6.7]	6.6 [6.3-6.9]
LVESD (cm)	1 month	5.3 [4.7-5.9]	6.1 [5.7-6.5]
	3 months	5.1 [4.6-5.7]	5.9 [5.5-6.2]
	6 months	5.0 [4.5-5.5]	5.7 [5.4-6.0]

Table 8. Changes of echocardiographic characteristics from baseline to 6 months.

Parameter	Δ MICS	Δ Sternotomy	p-value
Δ LVEDD (cm)	-0.9 [-1.2 to -0.6]	-0.7 [-1.0 to -0.4]	0.08
Δ LVESD (cm)	-1.4 [-1.9 to -1.0]	-0.8 [-1.3 to -0.5]	0.03
Δ RV FAC (%)	-2.4 [-3.5 to -1.2]	-1.1 [-2.6 to -0.4]	0.11

Table 9. Multivariable logistic regression for postoperative complications.

Variable	Odds Ratio (OR)	95% CI	p-value
MICS (vs sternotomy)	0.42	0.18-0.96	0.039
Age (per year)	1.02	0.98-1.06	0.31
BMI (per kg/m ²)	1.05	0.97-1.13	0.21
Prior cardiac surgery	1.41	0.52-3.82	0.50

morbidity associated with MICS was not solely driven by baseline differences in RV function. Sensitivity analysis stratified by baseline RV function is shown in Supplementary Table S3.

Discussion

In this retrospective comparative study based on a national cohort from Kazakhstan, we evaluated perioperative, early postoperative, functional, and echocardiographic outcomes following durable LVAD implantation using minimally invasive surgery and conventional median sternotomy. Several clinically relevant findings emerged. First, MICS was associated with a significantly shorter cardiopulmonary bypass time, although it did not result in lower intraoperative blood loss. Second, the composite burden of postoperative complications was significantly lower in the MICS group. Third, early recovery parameters, including intensive care unit and total hospital length of stay, were comparable between approaches. Fourth, survival and functional recovery did not differ significantly. Finally, both implantation strategies resulted in substantial reverse left ventricular remodeling, with a more pronounced reduction in left ventricular end-systolic diameter following MICS.

The association between minimally invasive LVAD implantation and reduced perioperative morbidity has

been consistently reported in contemporary literature. Meta-analyses and large reviews have demonstrated that lateral thoracotomy and other minimally invasive approaches are associated with reduced transfusion requirements, fewer reoperations for bleeding, and a lower need for right ventricular assist device implantation compared with full sternotomy (20-23). In particular, the systematic review and meta-analysis by Ribeiro et al. reported significant reductions in donor blood product use, reoperations for bleeding, and postoperative RVAD implantation following thoracotomy-based LVAD implantation (17). Our findings are concordant with this general trend, particularly regarding the overall composite complication burden.

In contrast to many published reports, we did not observe a reduction in intraoperative blood loss with MICS. This discrepancy likely reflects factors inherent to real-world implementation in an emerging LVAD program, including center-specific surgical techniques, the learning curve associated with minimally invasive access, and variability in blood loss documentation and hemostasis protocols. Importantly, CPB time was significantly shorter in the MICS group, an effect also reported by other centers and associated with reduced systemic inflammatory response and potentially lower complication rates (24). Prior studies suggest that with increasing experience, mini-access approaches may streamline operative workflow and reduce CPB duration (25,26), supporting the plausibility of our findings.

The reduction in the composite postoperative complication rate observed with MICS (30% vs. 50%, $p = 0.04$) represents one of the most clinically meaningful results of this study. This finding aligns with meta-analytic data demonstrating a lower overall complication burden and fewer reoperations for bleeding with minimally invasive approaches compared with sternotomy (27,28). While some studies have reported additional benefits such as shorter mechanical ventilation duration, reduced inotropic support, and shorter ICU stay, these effects appear to be center-dependent and influenced by local perioperative management strategies (29). In our cohort, the principal benefit of MICS was reflected in reduced composite morbidity rather than uniform improvement across all individual recovery parameters.

Functional outcomes, ventricular remodeling, and survival

Both surgical approaches were associated with marked functional improvement, with most surviving patients achieving New York Heart Association class II at 12 months. These results are consistent with outcomes reported for contemporary continuous-flow LVADs, particularly HeartMate 3. No cases of pump thrombosis or clinically significant hemolysis were observed, consistent with the favorable hemocompatibility profile of this device, although driveline infections remained the most frequent device-related complication irrespective of surgical approach (Table S4). Large registries and reviews report 1- to 2-year survival rates ranging from 75% to 90% for modern LVADs, with reduced rates of pump thrombosis and bleeding compared with earlier generations (30). A recent large-scale analysis demonstrated a 2-year survival rate of approximately 79% in contemporary HeartMate 3 cohorts (30), supporting the external validity of our findings.

Regarding ventricular remodeling, both implantation strategies resulted in effective left ventricular unloading and reverse remodeling. The greater reduction in LVESD observed in the MICS group is partially consistent with prior reports indicating that remodeling may occur independently of surgical access (31,32). This finding may also be influenced by more

favorable baseline right ventricular function among patients selected for MICS, supporting the concept that careful patient selection and preservation of right ventricular function may enhance unloading and remodeling responses (33).

Practical implications for resource-limited settings

In healthcare systems with limited access to heart transplantation, including Kazakhstan, durable LVAD therapy represents a cornerstone treatment for end-stage heart failure (34). Registry data demonstrate substantial international variability in LVAD utilization, transplant waitlisting, and transitions to transplantation, largely driven by donor availability and allocation policies (35,36). Evidence from the United States and Europe further indicates that successful transplantation following LVAD support varies widely across programs and over time (37,38). In this context, strategies that reduce postoperative morbidity and preserve end-organ function are particularly valuable, as they directly influence resource utilization, intensive care capacity, and long-term program sustainability.

A period-based analysis comparing early (2020-2022) and late (2023-2025) phases of the LVAD program (Tables S5-S6) demonstrated stable outcomes following minimally invasive implantation, suggesting procedural reproducibility despite increasing case volume. In contrast, outcomes following sternotomy varied between periods, particularly with respect to CPB duration, reflecting evolving surgical workflows and case complexity. These findings support the feasibility of minimally invasive LVAD implantation in emerging programs and indicate that its performance is not limited to the initial learning phase.

Limitations

This study has several limitations. The retrospective single-center design introduces potential selection bias and limits generalizability. The relatively small sample size reduces statistical power, particularly for infrequent complications and survival analyses. Some functional and laboratory parameters were available only at baseline or selected follow-up intervals, limiting

longitudinal assessment. Selection bias cannot be fully excluded, as patients with more preserved baseline RV FAC may have been preferentially selected for MICS. Finally, despite standardized surgical and perioperative protocols, unmeasured operator- and time-dependent factors may have influenced outcomes.

Conclusions

In this single-center national cohort, minimally invasive LVAD implantation was associated with a significantly lower composite postoperative morbidity burden and shorter cardiopulmonary bypass time compared with conventional median sternotomy, while providing comparable survival, functional recovery, and ventricular remodeling. Importantly, these advantages were demonstrated within an emerging mechanical circulatory support program rather than a high-volume referral center.

The value of minimally invasive LVAD implantation appears to lie not in improved survival, but in reducing early postoperative complications that directly affect resource utilization, intensive care demand, and program sustainability. In healthcare systems with limited access to heart transplantation, such morbidity-driven benefits are clinically and strategically meaningful. Collectively, these findings support the adoption of minimally invasive LVAD implantation as a default surgical strategy in carefully selected patients within developing LVAD programs. Further multicenter studies are warranted to confirm these results and to define standardized implementation pathways for resource-limited settings.

Ethic approval: The study was conducted in accordance with the principles of the Declaration of Helsinki and approved by the Local Ethics Committee of Kazakhstan's Medical University "KSPH" (protocol No. 4, approved on 21 May 2025).

Conflicts of interest: The authors declare no conflicts of interest.

Authors' contributions: B.T. and S.J. conceived the study and designed the research framework. S.Z. and U.K. developed the methodology. U.K. performed the statistical analysis and data processing. B.T., S.J., and A.B. conducted the investigation

and collected clinical data. Y.U., G.I., and A.B. provided study resources and institutional support. B.T., U.K., and A.B. curated the data. B.T., S.J., and S.Z. drafted the original manuscript. B.T., S.J., U.K., S.Z., Y.U., G.I., and A.B. critically reviewed and edited the manuscript. S.Z. prepared the figures and visualizations. Y.U. and G.I. supervised the study. Y.U., G.I., and A.B. managed the project administration. All authors meet the ICMJE criteria for authorship. All authors have read and approved the final version of the manuscript and agree to be accountable for all aspects of the work.

Statement on the Use of AI: Artificial intelligence-assisted tools were not used in the design of the study, data analysis, or preparation of this manuscript.

Informed Consent Statement: Given the retrospective design of the study and the use of anonymized routinely collected clinical data, the requirement for written informed consent for study participation was waived by the Ethics Committee. All patients had provided written informed consent for surgical intervention and the use of their clinical data at the time of hospitalization, in accordance with national regulations.

Data availability Statement: The data presented in this study are available from the corresponding author upon reasonable request. The data are not publicly available due to ethical and institutional restrictions.

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Appendix

Supplementary tables

Table S1. Detailed perioperative management and operative characteristics.

Parameter	MICS (n = 30)	Sternotomy (n = 30)
Peripheral CPB cannulation, n (%)	28 (93.3%)	0 (0%)
Central CPB cannulation, n (%)	2 (6.7%)	30 (100%)
Normothermic CPB, n (%)	30 (100%)	30 (100%)
Concomitant cardiac procedures, n (%)	4 (13.3%)	5 (16.7%)
Conversion to sternotomy, n (%)	0 (0%)	—

Table S2. Definition of composite postoperative morbidity endpoint.

Component	Definition
Reoperation for bleeding	Surgical re-exploration for bleeding during index hospitalization
Right ventricular dysfunction	Prolonged inotropic support or need for mechanical RV support
Neurological events	Ischemic stroke or transient ischemic attack
Acute kidney injury	Requirement for renal replacement therapy
Infection	Clinically significant infection requiring antibiotic therapy

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Table S3. Sensitivity analysis stratified by baseline right ventricular systolic function.

Baseline RV function	Approach	Patients (n)	Any complication, n (%)
RV $\geq$ median	MICS	15	4 (26.7%)
RV $\geq$ median	Sternotomy	14	6 (42.9%)
RV < median	MICS	15	5 (33.3%)
RV < median	Sternotomy	16	9 (56.3%)

Table S4. Device-related adverse events during follow-up (HeartMate 3).

Outcome	MICS (n = 30)	Sternotomy (n = 30)
Pump thrombosis	0 (0%)	0 (0%)
Clinically significant hemolysis	0 (0%)	0 (0%)
Driveline infection	3 (10.0%)	5 (16.7%)
Device malfunction	0 (0%)	0 (0%)
Pump exchange	0 (0%)	0 (0%)

Table S5. Temporal trends in LVAD implantation outcomes.

Period	Approach	Patients	CPB time, min (median)	Blood loss, mL
2020-2022	MICS	8	118.5	400
2020-2022	Sternotomy	10	95.0	240
2023-2025	MICS	22	120.0	400
2023-2025	Sternotomy	20	185.0	225

Table S6. Learning curve analysis within the MICS group.

Parameter	Early MICS (n = 8)	Late MICS (n = 22)
CPB time, min	118.5	120.0
Blood loss, mL	400	400